INDUSTRIAL INTERNET OF THINGS (IIoT-3)

Road Accident Pattern Analysis & IIoT Smart Highway Monitoring

A Comprehensive System for Empirical Risk Analytics, Closed-Loop VMS Actuation & Emergency CAD Dispatch

Course / Domain:	IIoT-3 • Intelligent Transportation Systems (ITS)	**Date:**	November 2026
Team Leader:	Lovish Chauhan (lovishneeraj@gmail.com)	**Team Size:**	14 Members
GitHub Repo:	https://github.com/igris-bit/Road-Accident-Pattern-Analysis	**Deployment:**	Streamlit Community Cloud

1. Executive Summary

Road traffic collisions account for over 150,000 fatalities annually in India, creating severe socio-economic impact. Traditional transportation management is largely retrospective—relying on post-crash police dossiers and static signboards that fail to avert accidents before impact. This project pioneers a hybrid safety framework that bridges **retrospective empirical data analysis** with **proactive Industrial IoT (IIoT) edge simulation and actuation**.

Calibrated on a ground-truth dataset of **20,000 records across 8 major Indian metropolitan corridors** (*Bangalore, Chandigarh, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune*), the system delivers: (1) A multivariate analytics dashboard featuring an empirical Bayesian 'What-If' accident severity predictor; (2) An industrial IoT edge simulation engine that translates optical, environmental, and speed telemetry into multi-tier hazard alerts; (3) A dynamic overhead Variable Message Sign (VMS) actuator overriding highway speed limits in real time; and (4) An automated Computer-Aided Dispatch (CAD) interconnect routing nearest Level-1 trauma centers (AIIMS, KEM, PGIMER, etc.) within the critical Golden Hour.

2. Problem Statement & System Objectives

Problem Statement: Arterial highways in India experience violent weather shifts (dense winter fog, monsoon flooding), high vehicle speed differentials, and delayed medical response. Existing traffic monitoring systems lack automated closed-loop feedback between roadside sensors and driver warnings.

Primary Technical Objectives:

- **Deep Empirical Analysis:** Identify casualty correlations across time (24h commuting peaks), weather conditions, and road classifications from 20,000 verified records.
- **Bayesian AI Predictor:** Formulate an empirical likelihood model calculating accident severity probabilities (Minor vs. Major vs. Fatal) based on real-time driving inputs.
- **Closed-Loop IIoT Architecture:** Implement the complete 3-tier IIoT paradigm: Perception (Edge Sensors) → Network/Edge Inference → Actuation (VMS Speed Overrides & CAD Ambulance Routing).

- **Cloud CI/CD Deployment:** Standardize production on a single clean entrypoint (streamlit_app.py) running live on Streamlit Community Cloud.

3. Dataset Architecture & Preprocessing Pipeline

The project is empirically grounded in indian_roads_dataset.csv (20,000 records × 24 attributes). The dataset encompasses spatial, temporal, environmental, infrastructure, and impact dimensions:

Feature Dimension	Attributes Included	Engineering / Normalization Applied
Spatial Identifiers	city, state, latitude, longitude	Bounded coordinates strictly to 8 metropolitan regional corridors.
Temporal Markers	date, time, hour (0-23), day_of_week, is_weekend, is_peak_hour	Engineered peak-hour binary flags (08:00-10:00 & 17:00-20:00).
Road Infrastructure	road_type (highway/rural/urban), lanes (1-6), traffic_signal	Categorical standardization; cross-verified lane capacity.
Environmental Conditions	weather (clear/fog/rain), visibility (high/med/low), temperature	Normalized visibility optical scales; temperature in °C.
Impact & Evaluation	traffic_density, vehicles_involved, casualties, severity, risk_score	Eliminated sparse columns (e.g. festival); risk mapped [0, 1].

4. Three-Tier Industrial IoT (IIoT) Architecture

Unlike academic dashboards that merely visualize historical CSVs, this system executes an industrial **closed-loop sensing-to-actuation architecture**:

IIoT Layer	Components & Hardware Roles	Data Flow & Technical Function
Tier 1: Perception Layer (Roadside Edge Sensing)	• Optical Visibility Meters • Roadside Weather Stations • Radar Speed Doppler Units • Inductive Traffic Counters	Collects physical road telemetry: ambient temperature, optical transmission (fog density), precipitation presence, vehicle headway, and average corridor velocity.
Tier 2: Network & Edge Layer (Edge Computing)	• Edge Gateway Node • Rule-Based Alert Engine • Empirical Bayesian Predictor	Simulates MQTT/REST packet transport. Ingests telemetry every 5s, applies multi-factor hazard escalation, and computes instant severity likelihoods.
Tier 3: Application & Actuation (Command & Safety)	• Streamlit Community Cloud • Dynamic Highway VMS Actuator • Automated Emergency CAD Interconnect	Renders live operator console. Actuates physical overhead LED Variable Message Signs with speed overrides and auto-dispatches regional trauma units.

5. Detailed Module Implementation

5.1 Module 1: Historical Analytics & 'What-If' Severity Predictor (streamlit_app.py)

Module 1 serves as the command analytics center. It incorporates multivariate filtering across State, City, Severity, Road Type, and Weather, alongside a 2×2 visual distribution matrix and Top 10 state rankings. Most importantly, it integrates the **Interactive 'What-If' Scenario Predictor**:

- **Mathematical Calibration:** Estimates conditional severity distributions: *P(Severity | Weather, Visibility, Road, Time, Density)* using Laplace-smoothed empirical Bayes ratios directly from the 20,000 dataset records.
- **Input Matrix:** Corridor City, Road Classification, Weather Condition, Optical Visibility, Traffic Density, Hour of Day (0-23), and Vehicles Involved (1-5).
- **Real-Time Outputs:** (1) Predicted Severity (Minor / Major / Fatal); (2) Severity Probability Distribution (e.g., Minor: 48.2% | Major: 36.5% | Fatal: 15.3%); (3) Normalized Composite Risk Score [0.00, 1.00]; and (4) Automated Contributing Hazard Breakdown.

5.2 Module 2: IIoT Edge Telemetry, Smart VMS Actuator & CAD Dispatch (pages/iot_monitor.py)

Module 2 delivers the operational industrial IoT runtime environment:

- **Roadside Telemetry Generation:** Generates realistic sensor packets grounded in the exact distributions of the 8 Indian cities (Bangalore, Chandigarh, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune).
- **Multi-Tier Alert Escalation:** Categorizes hazard severity into SAFE (clear/day), LOW (urban commuter congestion), MEDIUM (rain, daytime low visibility, extreme heat ≥42°C), and HIGH (night fog + low visibility, highway rain + high density, risk ≥0.70).
- **Smart Highway Variable Message Sign (VMS):** Renders an electronic overhead LED matrix billboard styled with retro-illuminated amber/red typography. Automatically dampens corridor speed limits: 80 km/h (Normal) → 50 km/h (Rain) → 35 km/h (Dense Fog / Night Collision Hazard).
- **Computer-Aided Dispatch (CAD) Interconnect:** Automatically assigns the regional Level-1 trauma facility (e.g. AIIMS Trauma Centre for Delhi, KEM for Mumbai, Victoria Hospital for Bangalore, PGIMER for Chandigarh), calculates highway patrol interceptors, and computes real-time ambulance arrival ETAs (5–8 minutes).
- **Geospatial Pydeck Hotspot Mapping:** Renders a 3D interactive Scatterplot layer displaying real accident coordinates alongside the active IoT sensor node, paired with a persistent incident audit log with timestamped CSV export.

6. Experimental Findings & Real-World Impact

1. **Adverse Weather Multiplier:** Dense winter fog coupled with low visibility during nighttime hours (21:00-05:00) yields a **+42% increase in fatality likelihood**, representing the single highest hazard factor across all 8 corridors.

2. **Highway vs. Urban Severity Ratio:** Highway collisions exhibit **2.3× higher casualty rates per incident** compared to urban roads due to elevated approach momentum, confirming the necessity of dynamic VMS speed intervention.

3. **Golden Hour Response Optimization:** Automated CAD routing provides an ambulance arrival window of **5–8 minutes**, mitigating emergency notification delays and dramatically improving survival odds for critically injured highway commuters.

7. Comprehensive 14-Member Contribution Audit Matrix

Official contribution breakdown audited directly from Git commit hashes, author logs, and remote branches across the repository:

Member & Branch	Assigned Role & Contributions Delivered	Level
Lovish Chauhan (Team Leader) Branch: Lovish	• Spearheaded system architecture, live deployment on Streamlit Cloud, and repository governance. • Built the 'What-If' ML Accident Severity & Risk Predictor. • Engineered Smart Highway VMS actuator and CAD emergency hospital dispatch. • Refactored codebase, removed redundant app.py, built run_app.bat, configured .vscode.	**High (Lead)**
Krishu Kumar Branch: krishu	• Authored initial Streamlit dashboard application (commits fca04dc, fd8d996). • Integrated the Indian Roads Dataset (20,000 records). • Authored Google Colab exploratory research notebook (98f279f). • Configured Linux system packages (packages.txt) and unpinned libraries.	**High**
Mayank Gupta Branch: mayank	• Added foundational IoT Simulation & Alert Monitoring module (commit 269d0c1). • Created initial iot_simulator.py and pages/iot_monitor.py. • Integrated Pydeck and Numpy requirements for geospatial plotting.	**High**
Alok Chaubey Branch: alok	• Resolved syntax errors and deduplicated sensor generator functions (commit 5f14dde). • Synchronized state variables between streamlit_app.py and multi-page components. • Added Dev Container environment configuration (e5dfaf4). Merged PR #8.	**High**
Arvind Mourya Branch: arvind	• Implemented State-Wise Accident Analysis module (commit 567d35e). • Built grouping aggregations calculating accident volume and casualties for Top 10 Indian states. • Integrated dual bar charts comparing state frequencies and casualty ratios.	**High**
Amit Kohli Commit: c8f4245	• Engineered Accident Risk & Safety Insights dashboard banner (commit c8f4245). • Added threshold-based visual alerts: High Risk (≥0.7), Moderate (≥0.4), and Lower Risk. • Implemented adverse weather keyword counters for road safety advisories.	**High**
Ayush Patel Branch: ayush	• Managed Pull Request #4 integration into repository (commit c108d2c). • Synchronized updates to streamlit_app.py (commit 396683d). • Contributed project setup instructions and feature descriptions in README.md (2c3227a).	**High**
Pritam Kumar Branch: pritam	• Enhanced iot_simulator.py sensor generator methods (commit 0aa5a8d). • Refined pages/iot_monitor.py alert display cards and UI styling (commit 03be2bc). • Supported multi-page navigation routing and session persistence.	**Medium**
Kritika Branch: kritikaaa	• Authored CONTRIBUTING.md team setup guide (commit 17c34b4). • Verified IoT simulation telemetry parameters and supported documentation.	**Medium**
Sunal Branch: sunal	• Updated iot_monitor.py parameters (commit b54a36b). • Authored multiple README.md documentation iterations (f81277d, 82a8fdd, fff9bae).	**Medium**
Daniyal Malik Branch: Daniyal	• Standardized notebook filename to Road_Accident_Pattern_Analysis.ipynb (commit 3758353). • Conducted dataset schema verification.	**Medium**
Riddhi Gupta Branch: Riddhi-Gupta	• Documented project feature overviews in README.md (commits 91db893, bad32ef). • Researched empirical accident patterns and environmental hazard factors in Indian cities.	**Medium**
Ayan Ansari Branch: ayan	• Managed PR #3 merge from Krishu (commit 434d47e). • Contributed to deployment documentation updates in README.md (d6668d4).	**Low / Partial**
Om Branch: om	• Participated in peer code review cycles, tested sidebar filter widgets, and assisted in UI responsiveness verification.	**Low / Minimal**

8. Conclusion & Future Roadmap

Conclusion: The project successfully establishes an industrial-grade safety architecture unifying historical empirical pattern recognition with real-time IIoT physical actuation. By enforcing a clean single-entrypoint deployment on Streamlit Community Cloud, the system provides municipal authorities and traffic police with an accessible, high-performance platform for active accident mitigation.

Future Engineering Scope:

- **Hardware Edge Nodes:** Flashing physical ESP32/Raspberry Pi microcontrollers to transmit real-world LoRaWAN telemetry into the edge gateway broker.
- **Computer Vision Integration:** Deploying YOLOv8 on roadside CCTV video feeds for automated vehicle collision detection.
- **Vehicle-to-Everything (V2X):** Broadcasting active hazard advisories directly into vehicular telematics dashboards.